Name: RISK 6.6 - Environmental harms

<Files\\Processed x Highly relevant\\2024-05-28 - WHC - AI and EVs add to soaring elec> - § 6 references coded [12.30% Coverage]

Reference 1 - 0.88% Coverage

â€œNot to mention the impact that AI will have on the electricity demand.â€

Reference 2 - 2.90% Coverage

â€œA really simple way to think about it is if you look at an average Google search and the energy consumption for that versus an AI Chat GPT search, the AI search requires 10 times as much power or energy as a regular Google search,â€ Taylor said.

Reference 3 - 2.09% Coverage

If we look at the United States as an example, data centres currently consume about 3 per cent of their overall power â€" by 2030 this was estimated to be 7 per cent, Taylor said.

Reference 4 - 1.23% Coverage

So estimates in the US say that over the next 10 years, they will need an extra 40 per cent more power.â€

Reference 5 - 1.28% Coverage

Taylor said that was putting discussion about more nuclear power back on the table in the Northern Hemisphere.

Reference 6 - 3.93% Coverage

â€œObviously to have data centres here is good for data sovereignty, but then we need the power to run them. Iâ€™m not sure that, in a short period of time, we could generate the additional electricity required for that, particularly if we want to move away from fossil fuels, because not all our power thatâ€™s generated is renewable.â€

<Files\\Processed x Highly relevant\\2024-12-12 - RNZ - Data centres ride AI boom in w> - § 4 references coded [13.74% Coverage]

Reference 1 - 3.70% Coverage

Training and using AI is an energy-hungry undertaking. Climate cost of AI in question (3 min 40 sec) According to Goldman Sachs, a ChatGPT query needs nearly 10 times as much electricity to process as a Google search and data centres globally are expected to more than double their power use by 2030. Reuters recently cited Morgan Stanley research putting data centres' 2030 climate impact at about 2.5 billion metric tons of carbon dioxide.

Reference 2 - 4.19% Coverage

But while the new wave of New Zealand centres are selling their sustainability, the average person typing in a query to ChatGPT or some other cloud-based programme is probably not putting an

energy load on a New Zealand-powered server - it is more likely in the United States.
Like the internet itself, the impacts of that energy use are global - unless a customer, such as the New Zealand government, has specifically contracted with the likes of Microsoft to have its data stored here in New Zealand.

Reference 3 - 1.53% Coverage

She says many of her contacts were horrified to learn how much water and energy their queries were using, when those resources could go to much more critical - even life-saving - uses.

Reference 4 - 4.31% Coverage

When I typed "data centre emissions" into an ordinary Google search bar to research this story, the first result was an AI generated overview from Gemini, Google's artificial intelligence chatbot - despite the query not having asked for an AI summary It turns out AI is well aware of its own impact, or at least of the servers on which it runs.
The summary said: "Data centres contribute to climate change by releasing greenhouse gases that cause rising temperatures, more severe weather events, and sea-level rise."

<Files\\Processed x Highly relevant\\2024-12-19 - NSM - Taking art from Artificial Int> - § 1 reference coded [0.97% Coverage]

Reference 1 - 0.97% Coverage

Could it lead to financial or environmental disaster?

<Files\\Processed x Highly relevant\\2025-02-12 - RNZ - New uses of artificial intelli> - § 1 reference coded [2.21% Coverage]

Reference 1 - 2.21% Coverage

Another constraint was power demand; the panel discussed how AI was incredibly hungry for power and water, and expected that would outstrip the country's energy supply in a decade or so.

<Files\\Processed x Highly relevant\\2025-02-28 - NZH - Outrage at university AI tutor> - § 1 reference coded [1.42% Coverage]

Reference 1 - 1.42% Coverage

â€œAI is constantly incorrect, environmentally damaging and such a stupid way of learning.

<Files\\Processed x Highly relevant\\2025-03-09 - SCP - AI Could Enhance Democracy - E> - § 1 reference coded [6.69% Coverage]

Reference 1 - 6.69% Coverage

The proposal was not supported by all councillors, with Crs Genevieve Robinson and Paul Dietsche raising concerns about cyber security and the amount of energy AI requires. Cr Robinson said AI has a reputation as an ""energy hog''. Cr Dietsche said the carbon footprint of AI ""is still to be dealt with''.